Two-Way Tables

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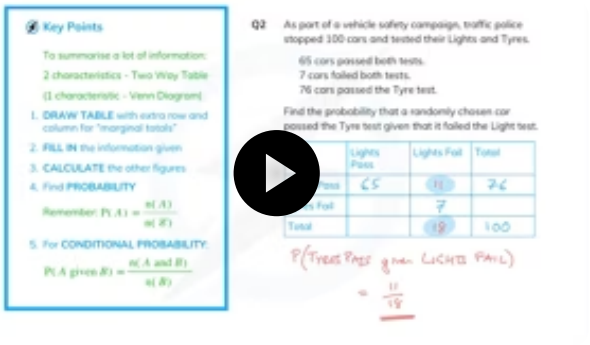

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Two-way tables

What are two-way tables?

- **Two-way tables** are tables that compare **two types** of **characteristics**
  - For example, a college of 55 students has two year groups (Year 12 and Year 13) and two language options (Spanish and German)
  - The two-way table is shown:

	Spanish	German
Year 12	15	10
Year 13	5	25

How do I find probabilities from a two-way table?

- Draw in the **totals** of each **row** and **column**
  - Include an **overall total** in the bottom-right corner
    - It should be the sum of the totals above, or to its left (both work)
  - For the example above:

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Got it

Year 12	15	10	25
Year 13	5	25	30
Total	*20*	*35*	*55*

- Use this to answer **probability** questions
  - If a random student is selected from the whole college, it will be out of 55
    - The probability a student selected from the college studies Spanish and is in Year 12 is $\frac{15}{55}$
    - The probability a student selected from the college studies Spanish is $\frac{20}{55}$
  - If a random student is selected from a **specific category**, the **denominator** will be that **category total**
    - The probability a student selected from Year 13 studies Spanish is $\frac{5}{30}$

Examiner Tips and Tricks

Check your row and column totals add up to the overall total, otherwise all your probabilities will be wrong!

Worked Example

At an art group, children are allowed to choose between colouring, painting, clay modelling and sketching.

A total of 60 children attend and are split into two classes: class A and class B.
12 of class A chose the activity colouring and 13 of class B chose clay modelling.
A total of 20 children chose painting and a total of 15 chose clay modelling.
8 of the 30 children in class A chose sketching, as did 4 children in class B.

(a) Construct a two-way table to show this information.

Answer:

Read through each sentence and fill in the numbers that are given

	Colouring	Painting	Clay modelling	Sketching	Total
Class A	12			8	30

Total		20	15		60

Use the row and column totals to fill in any obvious missing numbers

	Colouring	Painting	Clay modelling	Sketching	Total
Class A	12		15 - 13 = 2	8	30
Class B			13	4	60 - 30 = 30
Total		20	15	8 + 4 = 12	60

Use the row and column totals again to find the last few numbers

	Colouring	Painting	Clay modelling	Sketching	Total
Class A	12	30 - 12 - 2 - 8 = 8	2	8	30
Class B	30 - 12 - 13 - 4 = 1	20 - 8 = 12	13	4	30
Total	12 + 1 = 13	20	15	12	60

Write out your final answer

	Colouring	**Painting**	**Clay modelling**	**Sketching**	**Tota**
Class A	**12**	**8**	**2**	**8**	**30**
Class B	**1**	**12**	**13**	**4**	**30**
Total	**13**	**20**	**15**	**12**	**60**

(b) Find the probability that a randomly selected child

(i) chose colouring,

(ii) is in class A, who chose sketching.

Answer:

(i)

We are not interested in whether the child is in class A or B
A total of 13 children chose colouring, out of 60 children

$$\textbf{P(colouring)} = \frac{13}{60}$$

(ii)

8 children in class A chose sketching
There are 60 children to select from

(c) A child in class B is selected at random. Find the probability they chose painting.

Answer:

As we are only selecting from class B, this will be out of 30 (rather than the total of 60)
12 in class B chose painting

P(painting, from class B only) = $\frac{12}{30} = \frac{2}{5}$

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